Tri-Fusion Conceptual Whitepaper

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Tri-Fusion: A Three-Layer Framework for Transparent Information Synthesis
Mapping Narrative Structure Without Asserting Truth

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This work is published to establish prior art and encourage open research into transparent information synthesis systems.

Version 1

Abstract

Legacy and social media systems produce fragmented representations of truth due to algorithmic curation, biased framing, and narrative feedback loops. Current fact checking and media verification systems rely on centralized institutional authority or personal responsibility, often failing to holistically bridge societal trust divides. These approaches struggle because they require trust in institutions that users may reject, showing a need for a system which does not directly determine truth. **Tri-Fusion** introduces a three-layer framework that maps reported claims and generates an approximate lexically normalized narrative synthesis. The framework then audits the synthesis against observed source information. The framework **does not determine the truth**. However, it reveals the structure of reporting and sources' narrative consensus. This paper introduces conceptual architecture, **Tri-Fusion**, inviting open research, further iteration, and good-faith implementation. As Tri-Fusion is conceptual architecture, this paper does not present validated implementation. Included mathematical and pipeline models are proposed and theoretical for reference.

Critical Definitions

Claim

- A statement presented as factual by a source, regardless of its ultimate accuracy or intention
- Claims may be explicit or implicit, embedded in framing or tone

Narrative

- A structured collection of claims and framing used to describe an event

Narrative Landscape

- The full collection of narratives produced across all sources discussing an event

Claim Concurrency

- The proportion of sources within a narrative landscape reporting a specific claim

Narrative Divergence

- The degree of which narratives describing the same event differ in claim selection, framing, or writing structure

Narrative Diffusion

- The process of creating a summary of a narrative landscape, prioritizing structural normalization, cross referenced claims, and minimized lexical divergence

1.1. Introduction

Shared Reality and the Emergence of Narrative Fragmentation

The historical role of shared facts in discourse
Shared society-wide facts create the foundation for reasoning, discourse, and decision-making. Social norms, public policy, and scientific progress are dependent on verifiable and broadly agreed factual reference points. Algorithmic media creates an abundance of opportunities for divergence of shared reference points. As these reference points diverge, productive discourse becomes increasingly difficult.

The Modern Media Environment
Media ecosystems distribute information through algorithmically mediated channels which amplify engagement-driven content[4]. Different algorithms and socially segmented audiences produce divergent interpretations of shared events[5]. Engagement driven platforms critically increase recurring divergent interpretations through perpetuating slanted narratives[6]. Repeat exposure to truth-divergent claims may create false validity of claims through assertion[7].

Narrative Fragmentation
Narrative fragmentation occurs when interpretive divergence emerges at scale, altering the societal perception of shared events. This facilitates scenarios where large portions of societies see opposing narratives stemming from the same events. Through the facility of proof by assertion, productive discourse becomes mass polarization and in turn unproductive discussion. Narrative fragmentation may occur without intent: (accidental biases, epistemically charged wording, low quality depictions, etc) or with intent: (purposefully charged wording, altered depictions, skewed data, omission of information, algorithmic manipulation, etc).

Failure of Existing Approaches
Different approaches to solving problems created by narrative fragmentation exist. Centralized fact-checking employs case-by-case verification of claims by a single authority. Moderation tools are widely used to remove patently false or charged depictions of vetted information. Bias ratings are created for platforms which disseminate any media. These systems face key limitations when content volumes increase beyond the capacity of centralized verification[8].

Tri-Fusion's Purpose
Tri-Fusion is a three-layer architecture designed to map claims, synthesize narrative structure, and audit synthesized outputs. The framework facilitates transparent, user-driven analysis and verification of narrative structures, without central authority reliance. The first layer, **Cartographer,** creates a map of all related sources, claims, and relevant data within a specific narrative landscape. Layer two, **Diffuser,** contrasts every available source and data, with the help of **Cartographer**'s map. **Diffuser** then generates a **synthesized narrative** which attempts to normalize narrative divergences. **Auditor,** the final layer, compares all mapped claims and writing against the **synthesized narrative**, to examine and analyze gaps in claims and biases. **Auditor** also outputs an index measuring divergences in narrative landscape claims and lexical

structure. The three layers of **Tri-Fusion** compare narrative information across sources, evaluate a **synthesized narrative**, and identify structural biases within the narrative landscape. Tri-Fusion does not attempt to determine the objective truth of claims. Instead, it reveals patterns of reporting and narrative construction across sources.

Tri-Fusion analyzes how narratives are constructed and distributed across sources through three distinct analysis layers. The framework separates claim mapping, narrative diffusion, and audit reporting, with transparent and auditable design at each stage. Tri-Fusion does not claim authority over objectivity, instead functioning as a tool for user-driven interpretation of narrative structure and divergence. This framework is a conceptual contribution to narrative landscape analysis, inviting future iteration and implementation.

1.2. Scope and Non Goals

Limits in scope and potential function exist within the Tri-Fusion model. Tri-Fusion:

- Does not determine or claim authority over truth or objectivity
- Does not replace fact-checking, media literacy, or manual media analysis
- Does not guarantee unbiased or strictly objective outputs
- Is critically dependent on complete, accurate input

The remainder of this paper is structured as follows. Section two reviews specific existing systems addressing media biases. Section three outlines the epistemological ideas underlying the Tri-Fusion framework. Section four introduces the three-layer architecture, general implementation, and realistic applications. Section five discusses limitations, potential future research directions, and governance considerations.

2. Related Work and Existing Approaches

Before covering Tri-Fusion, this paper will first examine and define key failure points in current data concurrency and misinformation mitigation systems. These will serve as benchmarks for critical points of improvement for the Tri-Fusion framework.

Currently available tools to address media slanting and misinformation generally fall into four categories: source rating systems, independent fact-checking organizations, news aggregation platforms, and AI-driven summarization tools. All four approaches offer valuable tools for combating misinformation but have key structural limitations. These critical limits prevent each

tool from completely addressing narrative fragmentation.

2.1 Media Bias Rating Systems

Bias rating systems exist to give media platforms generalized ratings for bias, often updated on major world events or on a specific media release. Examples include AllSides, Media Bias Fact Check, and Ad Fontes. According to Ad Fontes Media Corporation[1], their ratings are created with a “team of political analysts” using Ad Fontes’ methodology. Ad Fontes’ system has the merits of professional, case by case analysis, with many chart and score style formats for ratings. This platform is limited by the lack of immediate availability. News cycles, events, and online trends can change hour to hour. Dependence on a team of analysts creates time bottlenecks between content publication and review. Additionally, the Ad Fontes hiring standards, data, and methodology breakdown are proprietary and hidden[1]. Media bias rating systems only apply to sources rather than individual stories and facts. Therefore, the bias within a given article or event’s full coverage may significantly diverge from the overall rating associated with the platform.

2.2 Fact-Checking Organizations

Certain organizations do specific fact checking independently or in service of a media platform. This includes organizations such as PolitiFact, Snopes, and FactCheck.org. For example, PolitiFact regularly fact-checks high profile political claims and posts for truth. This is done through an easily-digestible and rigidly defined truth-o-meter system[2]. Despite their strengths, these systems exhibit some key limitations. Distrustful audiences are quick to reject the truth-o-meter system as it is represented by a centralized authority, instead of repeated claim exposure[9]. Additionally, the system is limited by being reactive and not inherently systemically integrated. Individuals cannot put specific stories or claims into the truth-o-meter at rapid self-agency. Due to this reactive workflow, fact-checking organizations face scale constraints. Verification requires human research and analysis, critically limiting the amount of claims that may be analyzed at a given moment. Therefore, many underreported narratives, or those circulating only within social in-groups, remain unexamined.

2.3 News Aggregators

News aggregators use a broad spectrum approach to combat misinformation, slanted takes, and unintentional bias. They provide an individual with many sources for the same narrative, allowing an individual to draw a personalized and nuanced snapshot of claims based on the collection of sources. Services such as Ground News, Google News, and Apple News, are news aggregators with broad spectrum source functionality. Google News advertises itself as an easy gateway to personalizable news service. It comes with features like “full coverage”[3] which provides many different sources for the same event, alongside global source aggregation. While these features are very effective, they must be used as intended to generate a full picture of a given narrative landscape with the onus of processing placed on the individual users. Users may simply decline reporting from specific sources, or simply not want the burden of processing

every news story within a narrative landscape[10]. This dramatically reduces the positive outcomes of full coverage feature use. Additionally, news aggregators generally do not provide structural analysis for a particular story group, instead simply handling aggregation. Claim mapping, claim concurrency, and verification are also not handled by aggregation services.

2.4 AI and Large Language Model Summarization Tools

AI summarization helps everyday users navigate and verify high quantity news story landscapes. Mainstream AI models also come with safeguards and internet search capability, increasing quality of use for news summarization and understanding. Flaws with these tools do exist, particularly with non-locally deployed large language model systems. Their processing is done in a black box, without any insight into the decision making process of what a specific AI considers trustworthy and newsworthy[11]. Omission bias is also a major concern with AI use. In the process of maximal neutrality and factuality, AI systems may omit or ignore nuance and key quality issues behind data online and within its parameter set. This may alter the perception of a narrative through loss of emotional understanding, source quality, and omission of underreported claims. Another constraint is the collapse of source and interpretive plurality through AI processing. LLMs often collapse multiple narratives and writing styles into a neutralized summary. Disagreements and minority perspectives present in the wide narrative landscape are suppressed or removed from the summary. AI systems do not have accountability mechanisms for outputs and processing. Therefore, LLM aided news verification and understanding must always be taken with the understanding of opaque process and unaccountability[11].

All four approaches to addressing media biases and misinformation have specific, clearly identifiable constraints. These include: relying on a central authority, requiring manual interpretation, or opaque algorithmic processes. No approach completely provides a structured framework for system transparency, cross-source claim mapping, individualized at-agency use, and clear analysis of specific narrative landscapes. The Tri-Fusion framework proposal exists to address these gaps in a cohesive, complete manner.

3. Epistemological Foundations

Section three outlines the epistemological foundations which uphold the Tri-Fusion system's conceptual limitations, boundaries, and roles. As Tri-Fusion operates on large narrative datasets to perform media analysis, clear philosophical constraints are critical to preventing misinformation and misuse of this model. The following foundations lay clear limits of what Tri-Fusion is designed for and what it explicitly cannot do.

The Tri-Fusion framework's core philosophical foundations must be clearly laid out and documented to resist future misuse and misinterpretation.

3.1 Epistemic Humility

Tri-Fusion is a three-layer framework for mapping claims, synthesizing narrative consensus, and auditing synthesized narratives across multiple information sources. The Tri-Fusion architecture centers around epistemic humility as a foundational principle. The system cannot determine the truth of any claim or narrative outcome and is therefore not a verification authority[12]. Instead, Tri-Fusion uses large narrative landscape data to map: claim frequency, claim distribution, source distribution, narrative structure, and analysis. The framework functions as a tool to assist in interpreting complex narrative landscapes.

3.2 Observability Over Authority

Current information verification designs use opaque, centralized authorities to determine what claims are valid. The Tri-Fusion system rejects this approach. Instead, the framework prioritizes transparent observability. In this case, observability refers to the ability of users to personally inspect and evaluate information processing within the system. Every stage within Tri-Fusion's processing pipeline can be independently examined and audited by users.

3.3 Separation of Evidence and Trust

Empirical and normative signals occur frequently in applications of the Tri-Fusion framework. Empirical signals are extracted from observed patterns within the entire input narrative domain. Examples include claim frequency and narrative overlap within sources. Normative signals are evaluative judgements regarding sources. Examples include credibility ratings, trust scores, and user defined trust weights. The Tri-Fusion architecture maintains isolation between empirical and normative signals[14]. Empirical signals consist of observable media behavior, while normative signals consist of variable human judgements regarding credibility.
The system therefore avoids integrating empirical evidence and normative trust into a "truth score", to maintain both data streams' discrete integrity.

3.4 Pluralism of Interpretation

As Tri-Fusion does not assert authority over objectivity, its outputs remain open to user interpretation by design. The framework provides structured data analysis regarding reporting patterns, while final interpretation remains a human responsibility. This pluralistic design prevents Tri-Fusion from imposing or perpetuating a single authoritative narrative. Consensus does not imply truth or correctness, hence the importance of pluralistic interpretation[13].

In summation, these core principles establish the epistemological foundations and boundaries of the Tri-Fusion architecture and guide its responsible development and implementation.

4. Core Architecture of Tri-Fusion

Tri-Fusion's design uses a three-layer approach: Cartographer, Diffuser, Auditor. Each layer is critical to Tri-Fusion's function of claim mapping, synthetic narrative creation, and synthetic narrative auditing. See Figure 1 for a general layer process overview.

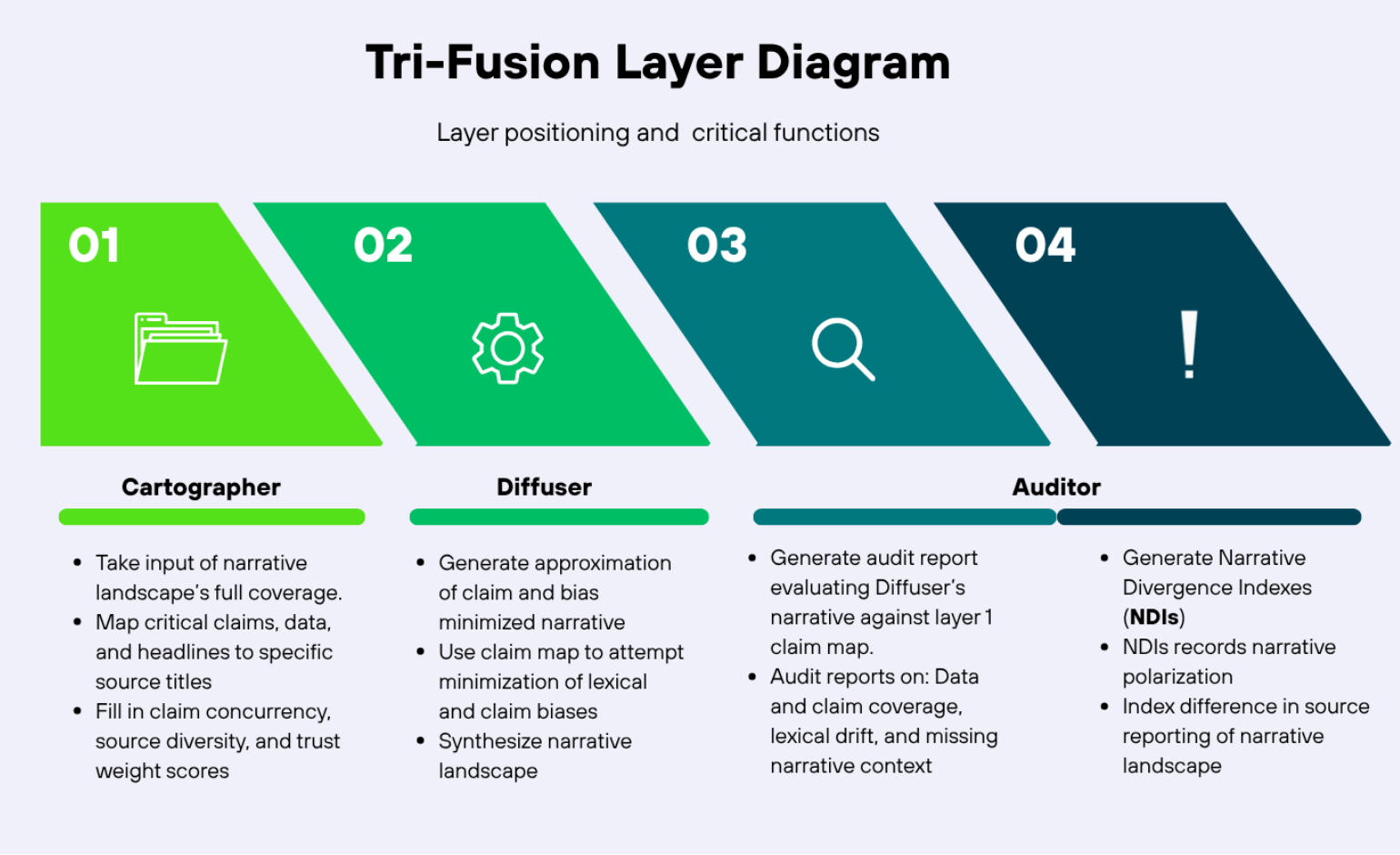

Figure 1, Layer function and order diagram of the Tri-Fusion Framework

4.1. Layer 1: Cartographer

Cartographer scans all sources within a media landscape and builds a map of all claims and data. This map links claims, metadata, full reporting, and headlines to an associated source. Per claim concurrency, source diversity and trust weights are also included in the claim map. *Cartographer does not filter, validate, or analyze claims, it only builds an observational map.*

Claim Registry

The claim registry contains all claims and critical claim information. Critical claim information includes raw-text claim representation, associated source, authors, and timestamp. This registry groups all claims with sufficient semantic similarity, based on embedding similarity. Similar claims are not identical, contradictory claims are separated. Claim registry accuracy is dependent on the breadth of available sources for a narrative landscape.

Concurrency Score

The concurrency score is an index examining the concurrency of a specific claim across all available sources. It is represented as a percentage of the sources covering an event reporting the claim. The concurrency score reveals claim reporting prevalence. It cannot take bias or framing into account for a claim's final score. Concurrency score does not represent the factuality of a claim, it only represents the rate of recurrence of a claim.

Source Attribute Weights

The source attribute weights give mathematical weightings to the specific political and reporting characteristics of sources. These weights cannot be internally sourced, they must be sourced from reputable external source rating or attribution systems. Source attributes describe characteristics of sources which may influence lexical structure and claim representation.

Source Diversity Index

The source diversity index measures the distribution of a given claim across source categories. Categories include: ownership, ideological distribution, geographic distribution. This index allows the observation of attributes which influence the probability of a claim being reported across different sources.

Trust Weights

Trust weighting defines the average reliability and accuracy of sources through evaluation. These weights cannot be sourced from within Tri-Fusion and must instead be manually input from external sources. Trust weights are normalized into a common scale and may vary by implementation. These weights add evaluative data-based context to every source's claims and

framing.

4.2. Layer 2: Diffuser

Diffuser creates a synthetic report for an event, by performing **Narrative Diffusion**. Narrative diffusion attempts to reduce divergences in lexical and structural framing across a narrative landscape. The structurally normalized result is then compiled into a synthetic narrative.

Key Objectives

Ordered Information Weighting
Diffuser finds narrative claims and data which have high narrative landscape concurrency, and relevancy, according to manually configured weightings. These data points are checked within the claim map and are prioritized for use in the synthetic narrative. Weighting influences prioritization but does not determine inclusion or factual accuracy. Inclusion is dependent on presence in claim map, meeting a narrative relevance threshold, and citation availability.

Low-Trust Information Compilation
Narrative claims with low concurrency scoring and parent source reliability are closely examined for usability. If claim map or external source corroboration is found for low-trust claims, they are incorporated into the final synthetic narrative.

Synthetic Narrative Creation
All final claims and framing are cross-referenced with the claim map. Diffuser processes framing, context, and relevant narrative structures. The synthetic narrative is then created with structurally consistent language, prioritizing data and claim framing mechanics over storytelling. Consistent, data oriented, and direct framing reduces framing-induced narrative fragmentation.

Key Constraints

Information Security
The narrative diffusion process cannot introduce new factual claims into the narrative landscape. Diffuser assumes sufficiently representative input to prevent introducing biases and incomplete information into the narrative diffusion process. Diffuser cannot guarantee full representation of source data and may reflect biases present in the claim map.

Framing Reduction
Narrative landscapes include source specific framing language to represent every claim and datapoint. Unique framing characteristics may produce subtextual biases, slants, or otherwise charged narrative delivery. The narrative diffusion concept must minimize framing into a simplified, data oriented delivery, to mitigate any potential framing biases.

Claim Citation

With every claim and datapoint output within the synthetic narrative, Diffuser must cite a source reference point from the claim map database. Citations ensure process accountability, cohesive narrative flow, and manual human review. Preventing obscure, unreliable, and unaccountable system design is a critical foundation for the Tri-Fusion framework.

Conceptual Model

Claim Selection

Diffuser first processes each source's claims for concurrency, trust, and critical source attributes. Claims are given higher priority upon corroboration in the claim map through multi-source presence and narrative landscape context. Low trust scoring does not eliminate inclusion in the synthetic narrative. It flags specific framing and claims to be corroborated or challenged with external sources.

Narrative Reconstruction

Selected claims are then compiled into a simple narrative reconstruction based on all sources' reporting. Full narrative reconstruction is then again cross-checked against the claim map, for citations and framing information. High divergences in semantic framing and claim citation are flagged as **high divergence**. Weights for calculating divergence, trust, and source attributes are manually configured to remove the onus of truth on Tri-Fusion.

Centroid Narrative Approximation

A centroid narrative is the minimum divergence point across the narrative reconstruction. This represents a weighted average of claim representations, not a consensus vote or objective authority. Diffuser enters an iteration cycle, where a copy of the narrative reconstruction is repeatedly iterated upon to create a centroid narrative. Narrative elements are broken down into specific vectors in semantic space, with **high divergence** flagged elements specifically highlighted. All narrative elements are then iteratively aligned in semantic space. High weight and concurrency elements have higher impacts on the centroid position. High divergence flagged elements receive extra scrutiny, to ensure they are reframed to align with the semantic position of the centroid narrative. Included outlier claims with low weighting contribute minimally to the centroid but are retained for full representation. The centroid narrative approximation is not representative of truth or neutrality. It only represents approximate statistical centrality within semantic space[16].

Synthetic Narrative Conversion

The centroid narrative, once exhibiting satisfactory element framing normalization, is rendered as readable human language. This pre-synthetic narrative is checked against the claim map and previous narrative reconstruction to ensure claim-map consistency, relevancy, and correct narrative flow. Previously high divergence flagged elements are re-scrutinized for semantic framing and are reframed as needed to match the centroid narrative.

Mathematical Model

Narrative Landscape: Collection of sources reporting on a given event. Mathematically represented by the claim map, a database of all sources, claims, and critical data for a specific narrative landscape.

Claims: A series of narrative elements written in-text within the claim map. Represented as vectors in high dimensional semantic space.

$$V_i = Embed(C_i)$$

Where: $Embed()$ is a function translating a discrete input into a high dimension vector[15]. This function is an approximation and may not fully capture semantic or contextual details.
V_i is a representation of $claim_i$ in vector space
Embeddings may lose context, distort polarity, and collapse lexical contradictions.

Each claim has a matching Citation and weight. Citations are a claim-specific pointer to claim map database entries containing a claim's source and context information. Weights are non-negative data points which affect the processing of each specific claim within the Narrative Diffusion process.

$$Weights\ w_i = f(concurrency_i, trust, source\ attributes_i)$$

Where: *Concurrency* is an index of matching claims
Trust is a normalized, manually input external score.
Source Attributes is a high dimension vector describing source-specific traits potentially affecting lexical framing and contextual use
f() is a configurable, implementation-dependent function to compute *Weights* w_i.

The Centroid Narrative Approximation
The centroid is the approximated minimum divergence point of all claims for a narrative landscape. It's influenced by claim semantic value, weight, and vector. This approximation is the crux of the Narrative Diffusion process. It minimizes weighted divergence across all claim vectors[16]. Embeddings are naturally lossy in contextual and structural preservation. Therefore, the centroid approximation is not a neutral truth, only a structural approximation at lexical equilibrium.

$$V_{centroid} = \frac{\sum_{i=1}^{n} w_i v_i}{\sum_{i=1}^{n} w_i}$$

This formula for computing the centroid narrative vector uses a weighted aggregation of every claim vector. The result is then normalized by all weights used. The end result is $V_{Centeroid}$, representing the minimized weighted divergence across all claim vectors.

4.3. Layer 3: Auditor

Auditor transparently examines the synthesized narrative based on all elements of the claim map and optional external cross-referencing data. It analyzes for omitted claims, trust alignments, lexical drift, missing context, and the **Narrative Divergence Indexes** (NDIs). These systems are designed to be manually examined in conjunction with the claim map and the synthesized narrative.

Audit Dimensions

Coverage

Auditor first processes which claims and datapoints were omitted from the synthesized narrative. Omitted claims are checked for source attributes, concurrency and trust scores. These attributes are then cross referenced with the narrative reconstruction, to see if critical information or points of view were lost due to the omitted claims.

Trust Alignment

High-trust and concurrency claim omissions may indicate issues with the narrative diffusion system. Auditor scrutinizes such omitted claims and examines the narrative context of the claims. The examined context includes source attributes, source information, narrative relevance, and similar claim inclusion. Auditor is dependent on the claim map's integrity and completeness. It cannot detect omissions outside the claim map, systemic omission across all sources, or coordinated narrative absence.

Lexical Drift

Framing from a centroid semantic vector is usually data oriented and reduces narrative fragmentation. Auditor processes the synthesized narrative for framing drifts and inconsistencies. If either inconsistencies or lexical drifts are found, auditor then checks the claim map for potential causes of centroid narrative approximation drift. This may include: lack of full narrative landscape representation, source collusion on preset narratives, or missing narrative elements in all claim-mapped sources.

Context Debt

Contextual debt occurs when key narrative, source, or contextual information is lacking from the claim map. This causes critical gaps and inaccuracies in the synthetic narrative. Auditor cross references the claim map and external sources to check for gaps in logical flow and source reporting. With these perspectives, Auditor suggests potential contributing factors for why context debt occurred. It also gives key data points and sources to include within the next use of Tri-Fusion, for a complete end result.

Output

Transparent audit reporting of layer one and two are Auditor's primary function. Auditor does not generate a corrected narrative or change its processed synthetic narrative. Reviewers are able to inspect key insights found within the synthetic narrative by Auditor. These include claim and contextual omissions, non-centroid framing, and claim uncertainties.

Narrative Divergence Indexes

Auditor's final, end of report outputs are the Narrative Divergence Indexes (NDIs). These indexes evaluate vocabulary and framing divergence across sources, synthetic narrative continuity, and claim inclusion rates. These statistics give descriptive, numerical insight to a narrative landscape's polarity, spectrum claim representation, and continuity with the synthetic narrative. Indexes are not raw computations or representations of truth. NDIs quantify divergence as the difference between narrative representations in semantic space.

Index Origin

Vocabulary and Framing Divergence Index: Examining claim map sources for net divergence in key attributes, vocabulary, and framing. Takes input of all claim and associated weight vectors, then computes embedded semantic divergence between all claims. Output index is normalized between 0 - 1.

Synthetic Narrative Continuity: Synthetic narrative continuity describes the synthetic narrative's integrity in reflecting the narrative landscape. Takes input of the non-centroid narrative reconstruction, compares narrative-critical claims represented in the synthetic narrative. Outputs a normalized inclusion rate 0-1. This does not identify fact from claims or lexical structure in the synthetic narrative.

Claim Inclusion Rate: This index presents a snapshot of total claims in a narrative landscape compared against claims included in the synthetic narrative. Takes input of all claims in claim map, then checks synthetic narrative for claim inclusion. Outputs a normalized, 0-1 index of claim inclusion.

5.1. Limitations and Failure Conditions

Limitations

Tri-Fusion has systematic limitations tied to the framework's design intent and operating conditions. These limits include individual implementation inconsistencies, output analysis and review divergences, and usage constraints. The listed limitations are included in Tri-Fusion by design and do not represent implementation issues.

Implementation Inconsistencies

The Tri-Fusion conceptual model is designed to accommodate individual design and development choices, to support further framework iteration and application. Specific applications of the conceptual model may have unique developments and ideas. However, they must adhere to the epistemological foundations and core technical model of Tri-Fusion. It is critical to represent Tri-Fusion as a tool to aid in centroid narrative landscape analysis.

Analysis and Review Divergences

Plurality of interpretation is a core design feature of Tri-Fusion. The onus of interpretation and understanding lies with the end user. Any application stemming from Tri-Fusion must not create authoritative narratives. Therefore, Tri-Fusion's applications must respect different human interpretations of its synthetic narrative and audit report.

Usage Constraints

Tri-Fusion is intended as a model for a tool to assist in narrative landscape interpretation. The model must always be used with the understanding that it is a tool with key constraints and failure modes. Truth, complete factual knowledge, and infallible reliability are not claimed or to be expected of Tri-Fusion. The system will perform as robustly as its implementation quality and user practices.

Limitations of Output Evaluation

Tri-Fusion's support for pluralistic evaluation creates limitations in evaluative robustness. Manual evaluation of results is difficult as no ground truth is explicitly represented in the model. Evaluation quality depends on input source completeness and diversity. Therefore manual evaluation results are descriptive and not correctness-based, providing consistent, input-dependent narrative analysis. Model outputs must be evaluated as functionally parallel to human summaries, having potentially incomplete claim coverage and non-objective structural alignment. NDIs must be leveraged to evaluate synthetic narrative divergence, inclusion, and continuity. Final manual evaluation measures narrative representation quality, being comparative, descriptive, and non-verificational.

Failure Conditions

Key framework specific failure modes exist within Tri-Fusion. These failures occur due to targeted information concealment, missing narrative landscape data, or narrative landscapes that have not sufficiently developed for effective analysis.

Systemic Blind Spots

Input sources are processed with the understanding of narrative landscape completeness. If sources omit or cannot report on specific key information, the system cannot reliably detect such omissions. Further, the Tri-Fusion model also cannot fill in or assume data for omission gaps. The framework can only highlight the potential existence of an omission within the auditing report.

Coordinated Misinformation

Sources may coordinate narratives and claims. Large scale coordination creates an inflated concurrency metric for Tri-Fusion. This may create low-trust claims and data with the appearance of high trust and repeat narrative presence[18]. The Narrative Diffusion process may be affected by large scale misinformation. This occurs when Diffuser synthesizes narratives assuming representative narrative landscape input. Auditor may report the existence of such abnormalities within the audit report.

Source Coverage Limits

Tri-Fusion cannot guarantee the input of a fully representative narrative landscape. Users must input sources for the landscape manually. Paywalled sources may be underrepresented due to missing concurrency and claim data points. Private or closed news systems may also be underrepresented due to lack of user access. This creates variations in narrative landscape accuracy, affecting the Narrative Diffusion process.

Early Event Uncertainty

Developing media landscape narratives often miss key narrative details regarding the specific event. Using immediate events as the narrative landscape of analysis may produce missing or otherwise incomplete results stemming from lack of story development. Immediate events are additionally prone to speculation and opinion-based reporting, which may further degrade the Narrative Diffusion process' quality.

Consensus Failure

Claims with high concurrency score may still reflect incomplete or incorrect narratives due to shared assumptions, biases, or untrustworthy fact pipelines. While Auditor attempts to mitigate this through its audit reporting and NDIs, the effect cannot be eliminated from the Tri-Fusion system. Therefore, consensus does not imply correctness, trust, or verifiability[17].

5.2. Open Framework Principles.

Open framework principles form a core bulwark for Tri-Fusion's identity. These principles establish foundational pillars for safe development of the Tri-Fusion framework

Transparency: Open design and development encourages transparency of internal systems. This allows for the development of better controlled and foundationally aligned applications of Tri-Fusion.

Auditability: Tri-Fusion preserves auditability for user and societal oversight. All steps of Tri-Fusion's model are able to be evaluated and inspected by users. Auditability prevents users from having blind faith in the model. A user may evaluate the model's processes to check for potential failure modes and issues, or for personal analysis.

Pluralistic Trust Models: Tri-Fusion defines metrics and systems to evaluate pluralistic trust. Sources within a media landscape are not simply trustworthy or untrustworthy. Every narrative element contains framing which may channel source attributes and create narrative fragmentation. Every output synthetic narrative and audit report support pluralistic interpretation to account for input source trust pluralism.

Institutional Capture Resistance:
Open framework principles allow for easy iteration and application of the Tri-Fusion framework. A diversity of applications creates variety in implementation and prevents single-institution centralization or dominance. Centralization presents a significant risk to Tri-Fusion's principles. Dominance may slowly create implied trust and may be perceived or presented as an authority for truth. Implied trust of Tri-Fusion's output eliminates pluralistic interpretation, a key safeguard in keeping Tri-Fusion as an analytical, non-authoritative system.

5.3. Governance

Open Specification

Tri-Fusion is an open specification framework. Independent and collaborative implementations and iterations are expected and encouraged. Reference versions of the original whitepaper are provided under appendix d when available, typically created and hosted by the whitepaper contributor.

Versioning and Iteration

New versions and iterations of Tri-Fusion are expected to reference the original Tri-Fusion

whitepaper. Future extensions and implementations are expected to be consistent with core principles covered in sections 3, 4, and 5. These cover the epistemological constraints, general architecture, and design limits of the Tri-Fusion framework.

Transparency and Non-Authoritative Constraint

Any further implementations must adhere to the open framework principles in section 5.2. The claim map, narrative diffusion process, and audit reports must be exposed and accessible to the user. No implementation should frame Tri-Fusion's outputs as truth or authoritative conclusions. Open framework principles must be upheld to mitigate the failure modes outlined in section 5.1.

Decentralization and Community Oversight

Iteration, implementation, and development of Tri-Fusion is highly encouraged. No external single, dominant entity should control or enforce the development and implementation of Tri-Fusion. The original whitepaper is cited as a reference for foundational principles and design intent. Future iterations must have methods for open and constructive critique, public evaluation, and academic review. Therefore, whitepaper-style paper iteration is encouraged.

5.4. Potential Applications

Tri-Fusion is an openly iterable framework, adaptable to many domains. The framework is especially suited for domains often interacting with narratives, claims, and information distribution. The listed potential applications maintain Tri-Fusion's non-authoritative and transparent design.

Media Analysis

Journalism and media analysis fields may benefit from tools like Tri-Fusion. The framework creates a synthetic narrative out of potentially hundreds of sources and articles. This improves analysis efficiency and remains transparent at the system level, for clear and accessible interpretation.

Academia

Journals and researchers may benefit from a Tri-Fusion style tool with additional Narrative Divergence Metrics and application specific adaptations. This enables bulk processing and research of media ecosystems, narrative divergences, and epistemology.

Public Information Systems

Public systems such as policy design and hearings may benefit from Tri-Fusion. The model allows for the efficient mass comparison of narrative and claim data, a traditionally time intensive process often occurring in public policy. Hearings feature framing reflecting personal

and political interests, which Tri-Fusion may be able to restructure into data-oriented analysis and clear data points.

Individual Use

Individuals may find utility in using the Tri-Fusion framework. The framework may provide value in personal media analysis, particularly in social and news media interactions. Tri-Fusion may also be a critical reading tool, assisting in narrative analysis and media literacy augmentation.

6. Conclusion

Tri-Fusion is a proposal which addresses the need for decentralized, transparent, on-demand narrative analysis. The framework provides narrative mapping, synthesis, and auditability for narrative landscapes. The framework does not attempt to determine truth or provide authoritative analysis of a media landscape. Tri-Fusion is guided by four foundational principles that define its behaviors and boundaries. The framework operates under epistemic humility, functioning strictly as an analytical tool without verification authority. Tri-Fusion includes observability, allowing all layers of the framework to be inspected and evaluated. Separation of evidence and trust is key to Tri-Fusion's processes by keeping normative and empirical source data separate in processing, reducing the risk of data integrity loss. Tri-Fusion additionally supports pluralistic interpretation, preventing the framework from imposing authoritative interpretations. In doing so, it shifts the role of artificial systems from authorities of truth to instruments of epistemic analysis and transparency. By exposing narrative structure instead of asserting authority over truth, Tri-Fusion aims to support more informed reasoning in complex media environments. This framework is intended to contribute a foundation of future research, implementation, and development to the domain of narrative analysis.

Appendices

Appendix A — Glossary

Cartographer: Tri-Fusion's first layer. Creates relational maps of claims, claim concurrency, sources, and source attributes across a narrative landscape.

Diffuser: Tri-Fusion's second layer. Takes input of claim map, manual trust and source attribute weightings. Uses embedding functions to approximate a centroid narrative. Synthesizes media landscape description, leveraging key claims and weightings, approximately aligned with the centroid narrative.

Auditor: Tri-Fusion's final layer. Analyzes manually input external sources and narrative landscape claim map, against synthetic narrative. Attempts to highlight narrative

inconsistencies, underrepresented claims, and logical or factual inaccuracies. Generates Narrative Divergence Indexes based on key data points. NDIs give approximate indexed values for narrative polarization, framing and vocabulary divergence, and a ratio of the claims in the claim map not represented in the synthetic narrative.

Narrative Diffusion: The process of creating a synthetic narrative through narrative reconstruction, centroid approximation, and narrative synthesis. Occurs in the Diffuser stage of Tri-Fusion.
Narrative Divergence: Differences in lexical representation of claims, narrative structure, and opinion in a discrete narrative landscape.

Appendix B — Minified Tri-Fusion Process Example

This appendix provides a simplified end-to-end example of Tri-Fusion's three-layer architecture. It is not a representation of how the system may function at further implementation or iteration. Values and functionality are illustrative and are not intended to articulate complete function.

1: Claim Map Example

Source	**Claim Tuple**	**Weight** (concurrency, trust, source attributes[vector])	**Claim Contextual Quote**	**Claim Citation**
A-News Network	**1 Boy Injured**	(3, 48, a)	"... one boy in critical condition after potential zzzSystems gate failure, corporate internal probe underway…"	Source Index: [1] Media Index: [4] Claim Page: [9] Claim Tuple Starting Character Position: 30041 Contextual Quote Starting Character Position: 29910
B- Routers Network	**1 Youth Injured**	(3, 67, b)	**"…** accident at Cron LRT station, one youth seriously injured, authorities investigating…"	Source Index: [2] Media Index: [12] Claim Page: [8] Claim Tuple Starting Character Position: 97603 Contextual Quote Starting Character Position: 97547

C-Truth Television	**Citizen Seriously Injured**	(3, 29, c)	**"...**corporate accountability needed after the critical injury of a New Arcadia citizen from faulty zzzSystems LRT gate design…**"**	Source Index: [3] Media Index: [2] Claim Page: [13] Claim Tuple Starting Character Position: 19567 Contextual Quote Starting Character Position: 19471

Reference Source Attribute Vectors
a = (1,6,−23,0,0,400)
b= (9,12, 45,19,3,-800)
c= (2,40,78, 27,11,900)

2: Single Claim Narrative Synthesis Example

Claim Definitions	**Centroid Approximation**	**Synthesized Narrative**
Claim c_1 = "1 Boy Injured" Weights w_1 = f(3, 48, a)	$$V_i = \frac{\sum_{i=1}^{n} w_i c_i}{\sum_{i=1}^{n} w_i}$$	One youth seriously injured after LRT gate malfunction, authorities investigating.
Claim c_2 = "1 Youth Injured" Weights w_2 = f(3, 67, b)		
Claim c_3 = "Citizen Seriously Injured" Weights w_3 = f(3, 29, c)		

3: Single Claim Audit Example

zzzSystems' potential situational responsibility as faulty gate's designers is not mentioned. (Citations: [3:2:13:19567:19471], [1:4:9:30041:29910]).

Victim's New Arcadian nationality not divulged. (Citation: [3:2:13:19567:19471])

Location of event at Cron LRT station not given. (Citation: [2:12:8:30041:29910])

Appendix C — Future Research

Future research and iteration will benefit Tri-Fusion's robustness as an implementable framework. Key research domains include: narrative vector modeling, diffusion algorithm engineering, system adversarial resilience integration, and further development of divergence metrics.

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